

MAT101 PRESENTATION

Topic: Integration

Team Members:

- 1. Abdullah Al Junaid - 2024-3-60-082**
- 2. Fazle Rabbi - 2024-3-60-084**
- 3. Nausaba Alam - 2024-3-60-099**
- 4. Al Nahian Adipto - 2024-3-60-101**
- 5. Samia Zaman 2024-3-60-145**



In this presentation, we will discuss real-life applications of integration. Integration plays a significant role across various fields.

Some of these applications include:

- Time Complexity Calculation
- Kinematics
- Cumulative Distribution Functions
- Continuous Probability Distribution
- Volume of Solid Revolution



1. Time Complexity

Time complexity measures how long an algorithm takes to run as a function of the input size (n)

Importancy:

- **Optimizing Code:** Identifies inefficiencies for better performance.
- **Scalability:** Ensures algorithms handle larger inputs effectively.
- **Comparison:** Provides a standard way to compare algorithm efficiency.

Example:

- **$O(1)$:** Constant time
- **$O(\log n)$:** Logarithmic time etc.



Rules of Big O:

Big O focuses on

- The worst-case scenario
- Ignoring constant factors

Now, let's see some application of integration to understand how we can calculate Big O notation.



Linear Time Complexity:

Mathematical Analysis with Integration:

$$T(n) = \sum_{i=1}^n 1$$

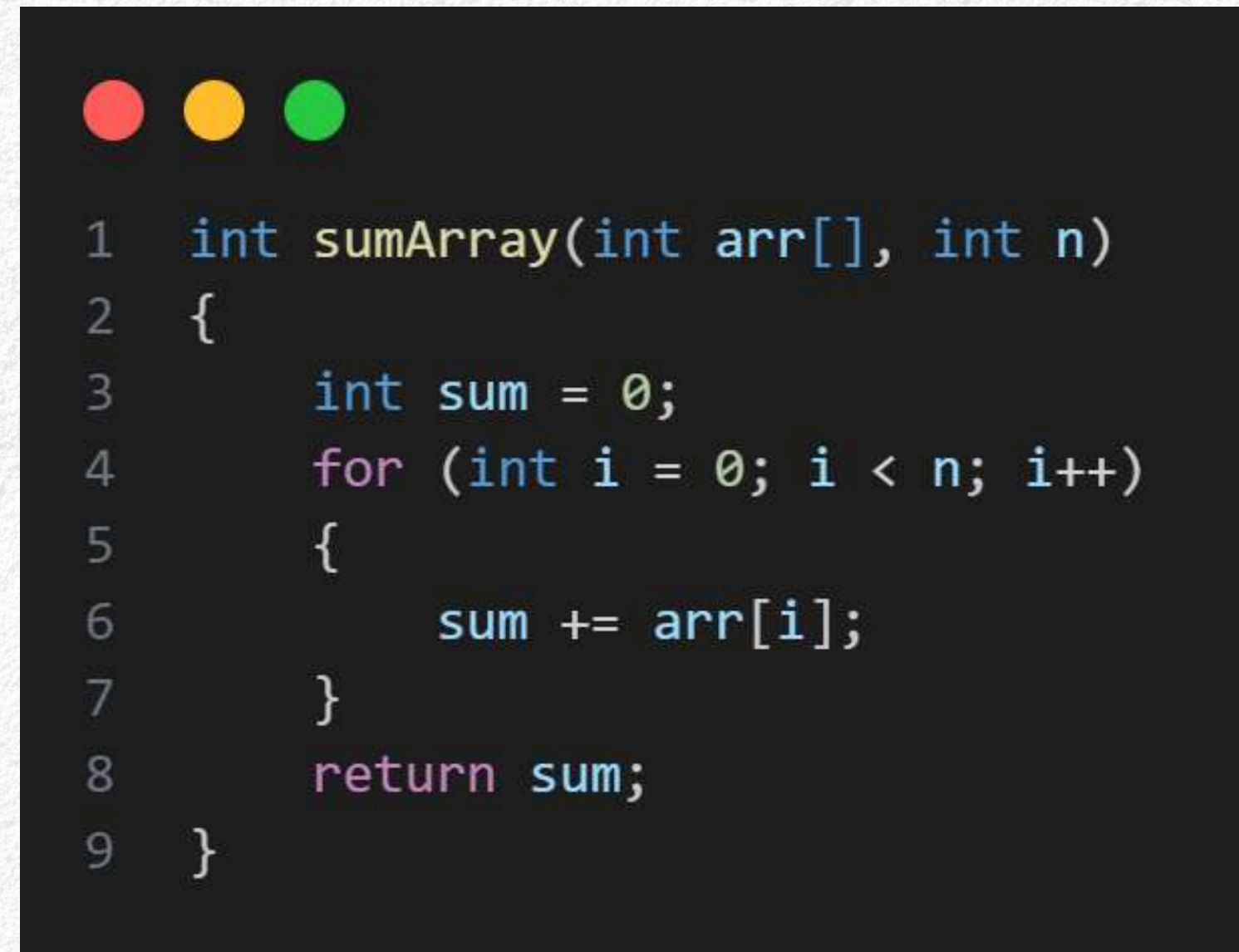
Convert the summation to an integral:

$$T(n) = \int_1^n 1 \, dx$$

$$= [x]_1^n$$

$$= n - 1$$

Thus $T(n) = O(n)$



```
1  int sumArray(int arr[], int n)
2  {
3      int sum = 0;
4      for (int i = 0; i < n; i++)
5      {
6          sum += arr[i];
7      }
8      return sum;
9  }
```



Linearithmic Time Complexity:

Mathematical Analysis with Integration:

$$T(n) = \sum_{i=0}^{\log n} n$$

Convert the summation to an integral:

$$\begin{aligned} T(n) &= \int_1^{\log n} n \, dx \\ &= n \cdot (\log n) - n \cdot (1) \end{aligned}$$

Thus $T(n) = O(\log n)$

```
1  #include <stdio.h>
2  int main()
3  {
4      for (int i = 1; i < n; i++)
5      {
6          for (int j = 1; j < n; j = j * 2)
7          {
8              printf("%d %d\n", i, j);
9          }
10     }
11 }
```



Exponential Time Complexity:

Mathematical Analysis with Integration:

$$T(n) = \sum_{i=1}^n 2^i$$

Convert the summation to an integral:

$$T(n) = \int_1^n 2^x dx$$

$$= \left[\frac{2^x}{\ln 2} \right]_1^n$$

$$= \frac{2^n}{\ln 2} - \frac{2}{\ln 2}$$

Thus $T(n) = O(2^n)$

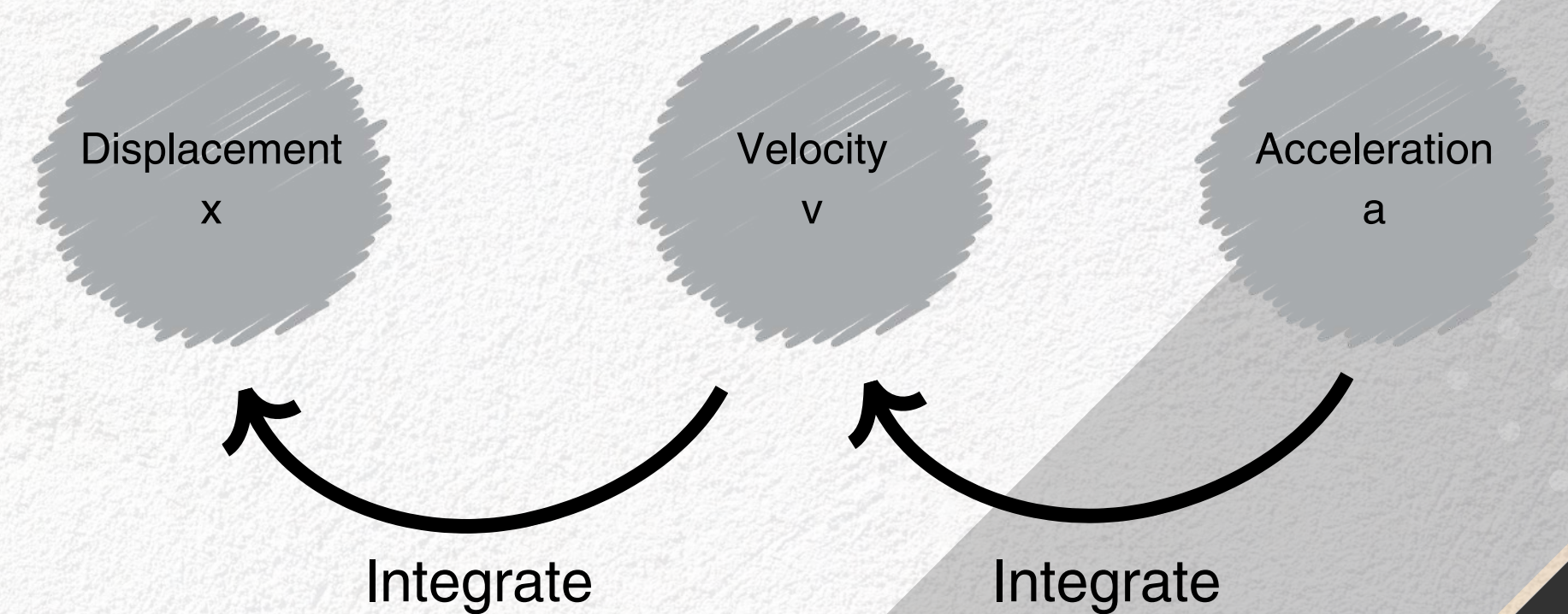


```
1  int fibonacci(int n)
2  {
3      if (n <= 1)
4          return n;
5      return fibonacci(n - 1) + fibonacci(n - 2);
6  }
```



2. Kinematics

- Integration in kinematics is a powerful mathematical tool used to analyse motion; particularly when dealing with quantities that change over time like velocity, acceleration and displacement



Finding displacement from velocity

The relationship between velocity and displacement is given by

$$s(t) = \int v(t) dt + c$$

where:

- $s(t)$ is the displacement as a function of time.
- $v(t)$ is the velocity as a function of time.
- c is the constant of integration, which represents the initial position $s(0)$.

Example: If a particle has a velocity $v(t) = 3t^2$, its displacement is: $s(t) = \int 3t^2 dt = t^3 + c$



Finding Velocity from Acceleration

The relationship between acceleration and velocity is given by:

$$v(t) = \int a(t) dt + c$$

where:

- $a(t)$ is the acceleration as a function of time.
- c is the constant of integration, representing the initial velocity $v(0)$.

Example: if $a(t) = 4t$, then:

$$v(t) = \int 4t \, dt = 2t^2 + c$$



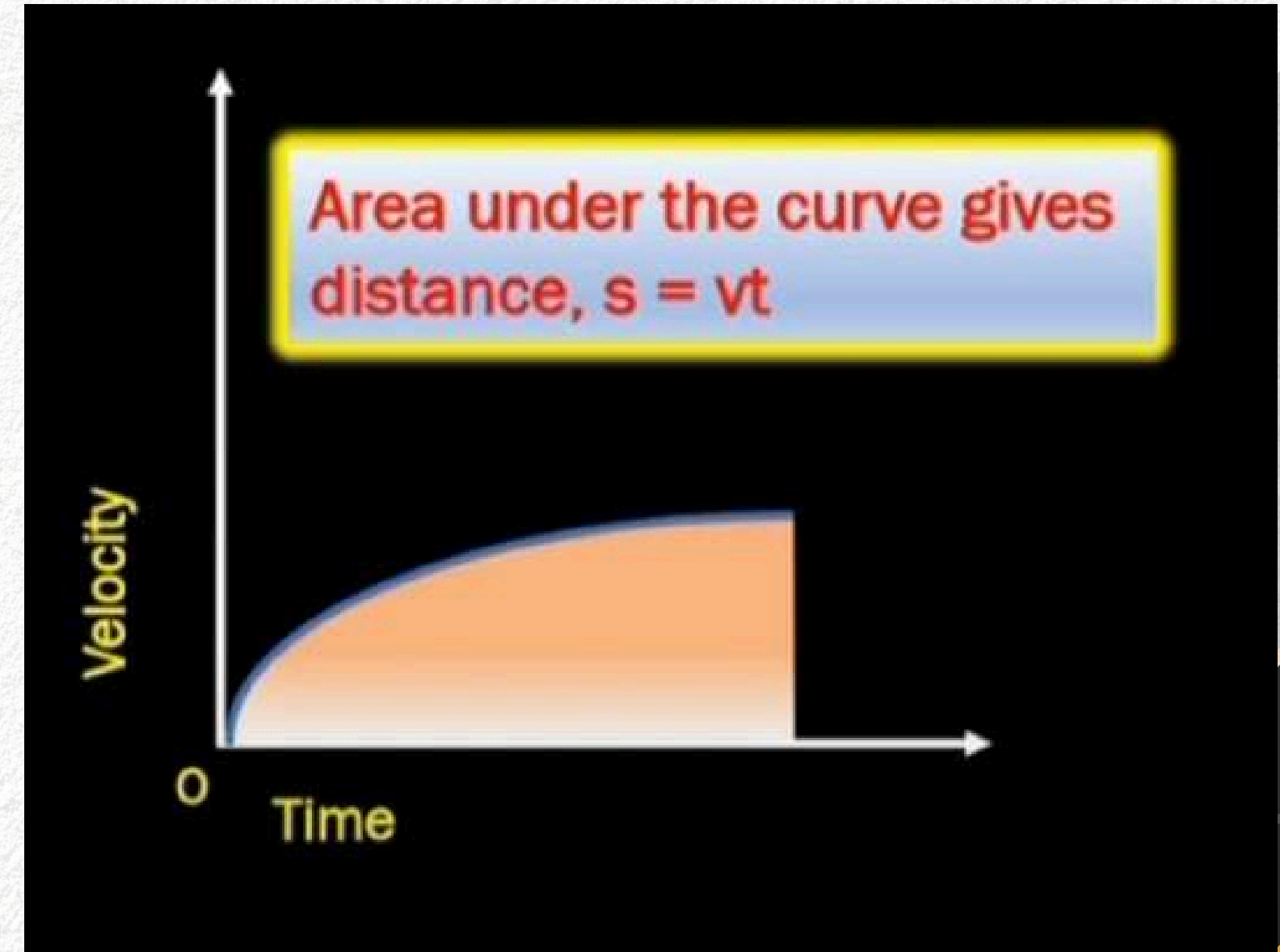
Calculating total distance travelled

When velocity changes sign, displacement alone may not give the total distance travelled.

The total distance can be calculated as:

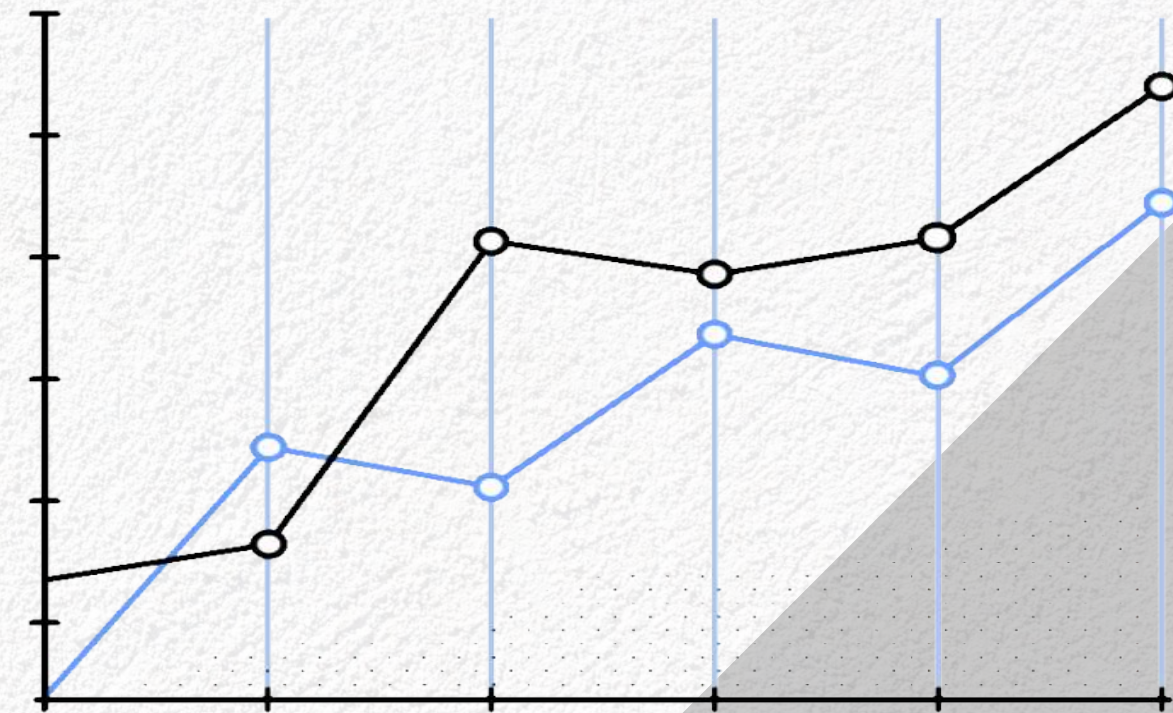
$$\text{Total distance} = \int |v(t)| dt$$

Here, $|v(t)|$ ensures that the magnitude of velocity is considered, regardless of direction.



3. Econometrics:

Integration plays a fundamental role in econometrics, particularly in the analysis of time series data, probability distributions, and the evaluation of models.



Cumulative Distribution Functions:

Integration is used to derive CDF
from probability density function
PDF

If $f(x)$ is the PDF of a variable x , the
CDF $F(x)$ is,

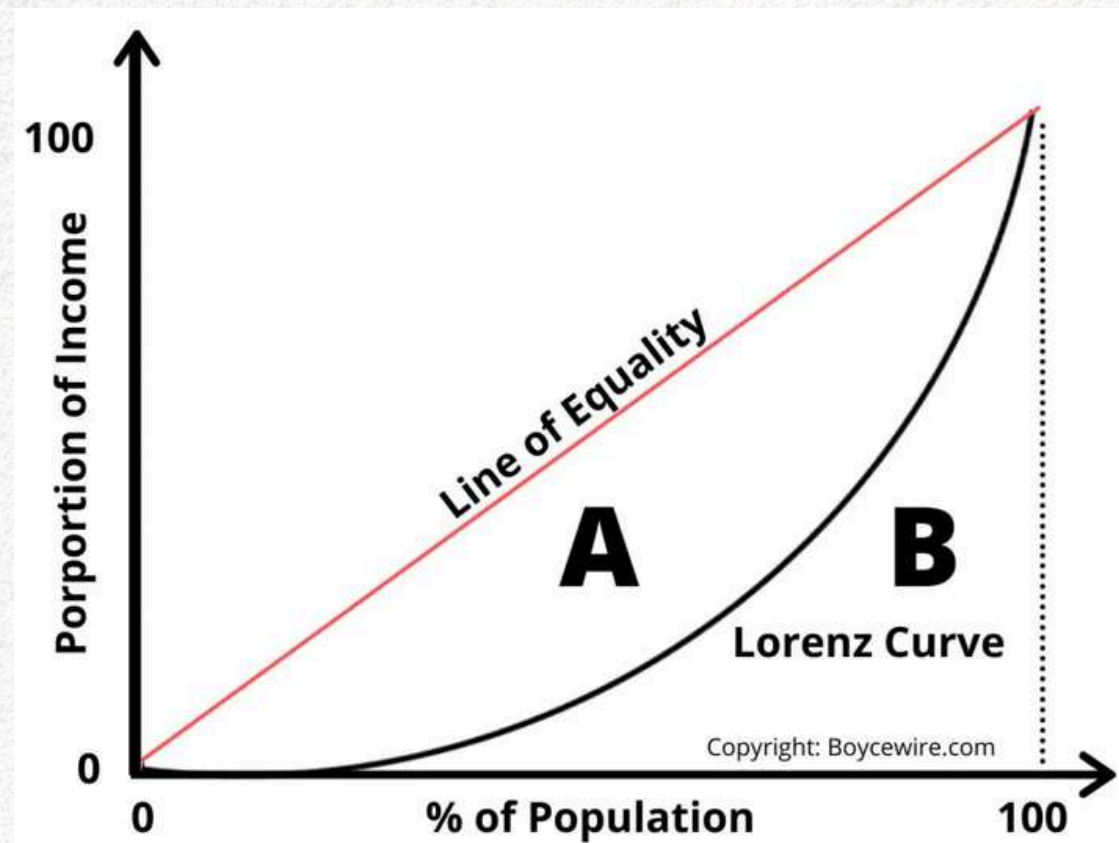
$$F(x) = \int_{-\infty}^x f(x) dx$$

The average outcome of a variable x
is $P(x) = \int_{-\infty}^{\infty} x f(x) dx$



Lorenz Curve:

Lorenz curve is a graph of income inequality that shows what percentage of a country's income is being earned by a percentage of the country's households.



$$\text{Gini Index} = \frac{A}{A+B}$$

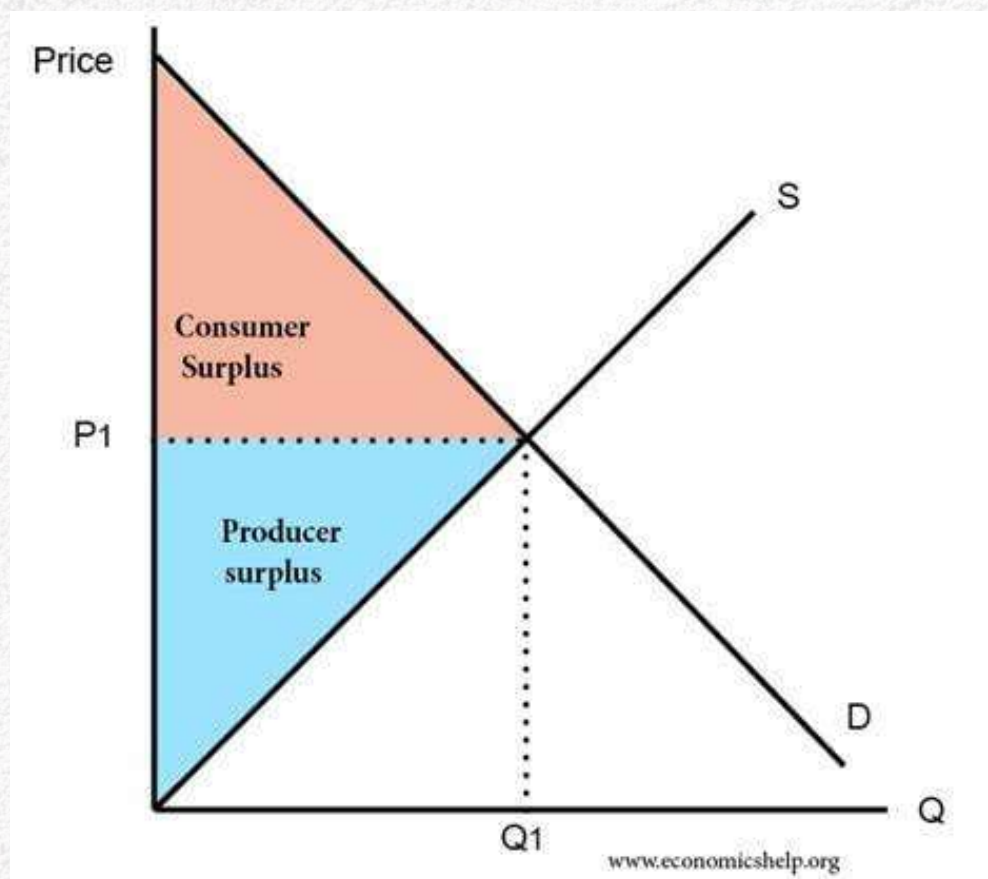
$$\text{Gini index} = 2 \int_0^1 [x - f(x)] dx$$

where $f(x)$ is the lorenz curve.



Total Surplus:

Total surplus is the sum of consumer surplus and producer surplus. It represents the total welfare or economic benefit to society from trade the market



Total surplus = Consumer Surplus + Producer Surplus

$$= \left[\int_0^{Q^*} D(Q) dQ \right] - \left[\int_0^{Q^*} S(Q) dQ \right]$$



$D(Q)$ = Price consumer are willing to pay at Quantity Q .

Q^* = Quantity sold in equilibrium.

$S(Q)$ = Price producers are willing to accept at quantity Q .

4. Continuous Probability Distribution

A probability distribution where the random variable can take any value within a certain range. For example: normal distribution, exponential distribution, uniform distribution.

How integration plays a role:

If X is a continuous random variable with probability density function $f(x)$, then the expected value of X , denoted as $E(X)$, is given by,

$$\int_{-\infty}^{\infty} x \cdot f(x) dx$$



Let's say,

For a random variable X , the probability distribution function denotes

$f(x) = \int_{-1}^1 x \cdot \frac{1}{\pi \sqrt{1-x^2}} dx$ for $x \in [-1,1]$. Now, to calculate the expected value of

this distribution we need to compute the integral of form.

So, $E[X] = \int_{-1}^1 x \cdot \frac{1}{\pi \sqrt{1-x^2}} dx$

{note: using the formula of probability distribution}



(Method: Integration by substitution)

$$= \frac{1}{\pi} \int_{-1}^1 \frac{1}{\pi \sqrt{1-x^2}} dx$$

$$= \frac{1}{\pi} \cdot \frac{1}{-2} \int_0^1 \frac{1}{\sqrt{u}} du \quad [\text{considering, } u = 1 - x^2 \text{ or, } du = -2x dx]$$

$$= \frac{1}{\pi} \cdot \frac{1}{-2} \frac{u^{-1/2+1}}{-1/2+1}$$

$$= \frac{-1}{\pi} [\sqrt{u}]_0^1$$

$$= \frac{-1}{\pi} [\sqrt{1-x^2}]_{-1}^1$$

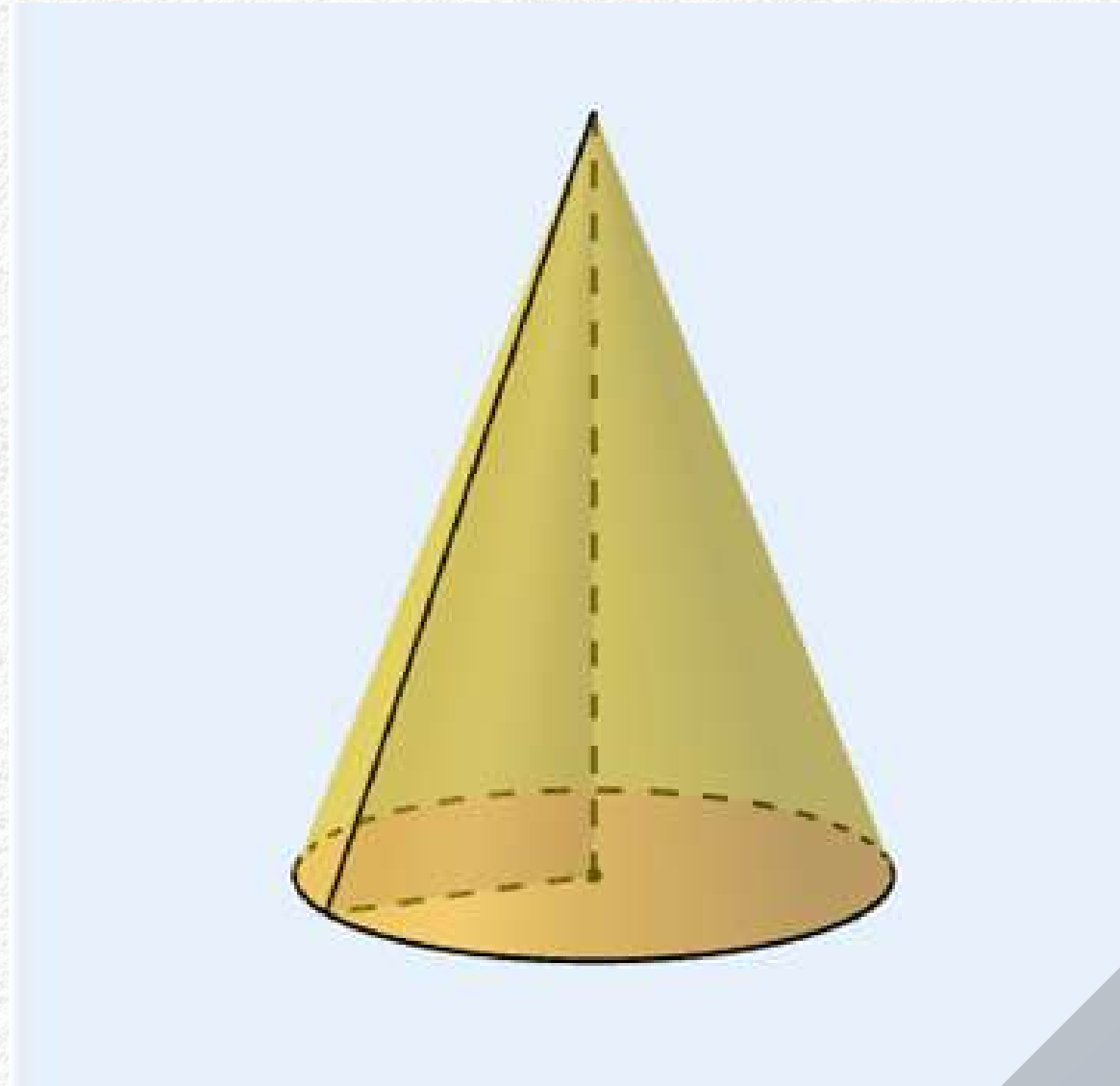
$$= \frac{-1}{\pi} + \frac{1}{\pi}$$

$$= 0$$

The expected value provides a useful tool for predicting average outcomes in uncertain situations. Which was done by using the laws integration.

5. Volume of Solid Revolution

A solid of revolution is a three-dimensional shape created by spinning a two-dimensional curve around a line within the same plane. The volume of such a solid can be computed using integration. The most typical techniques for determining the volume include the disc method, the shell method, and Washer Method



Disk Method:

Disk Method is the method of finding the volume of the 3-D figure in this method, we take axis of revolution the boundary of the plane region and the cross sections are taken perpendicular to the axis of the revolution. Now the cross section area of the disk is the πr^2 and the volume of the disk is the area times its thickness.

- If the disk is perpendicular to the x-axis then its radius is the function of x.

$$V = \int_b^a \pi [f(x)]^2 dx$$

- If the disk is perpendicular to the y-axis then its radius is the function of y.

$$V = \int_b^a \pi [f(y)]^2 dy$$

Example:

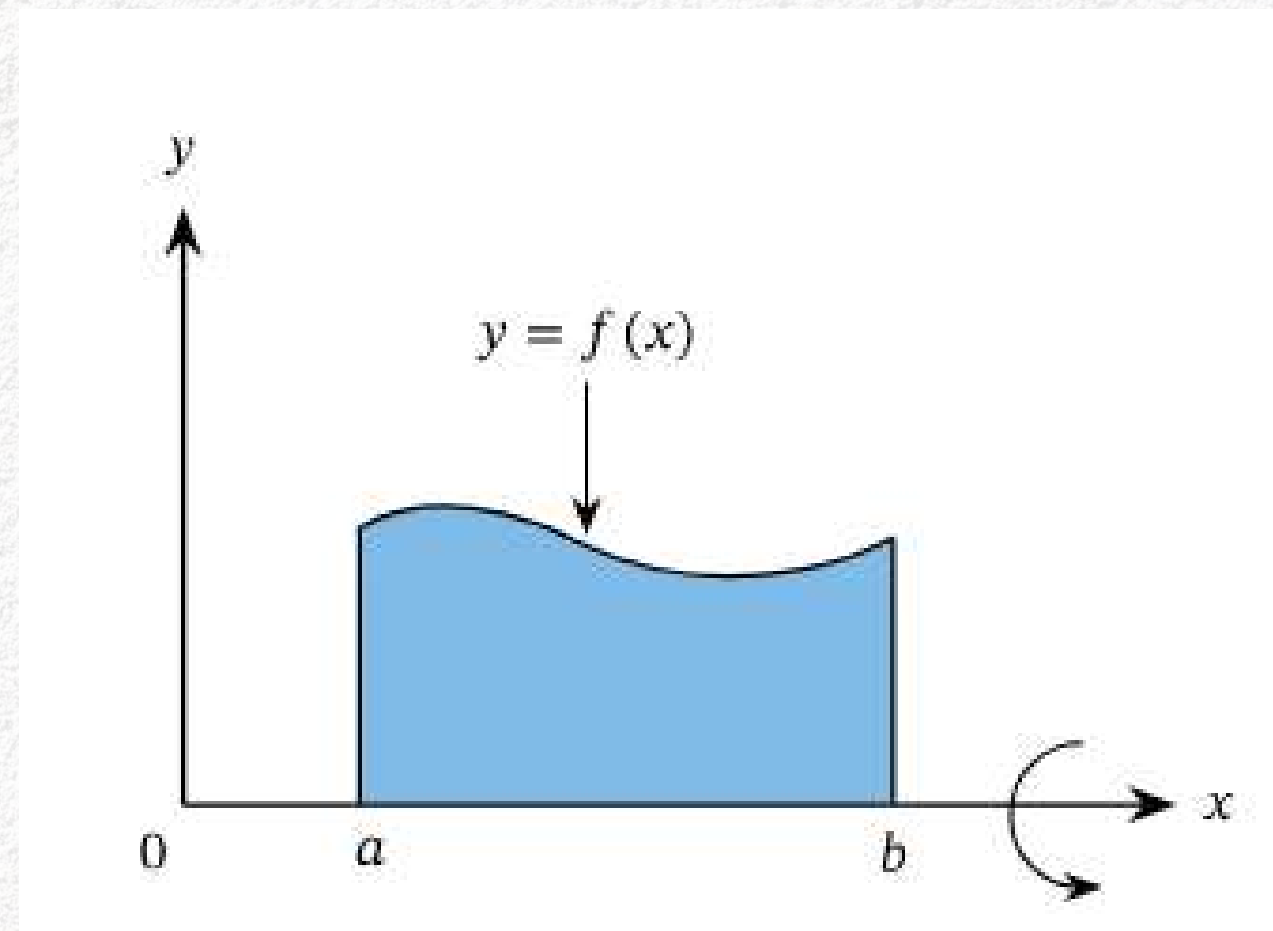
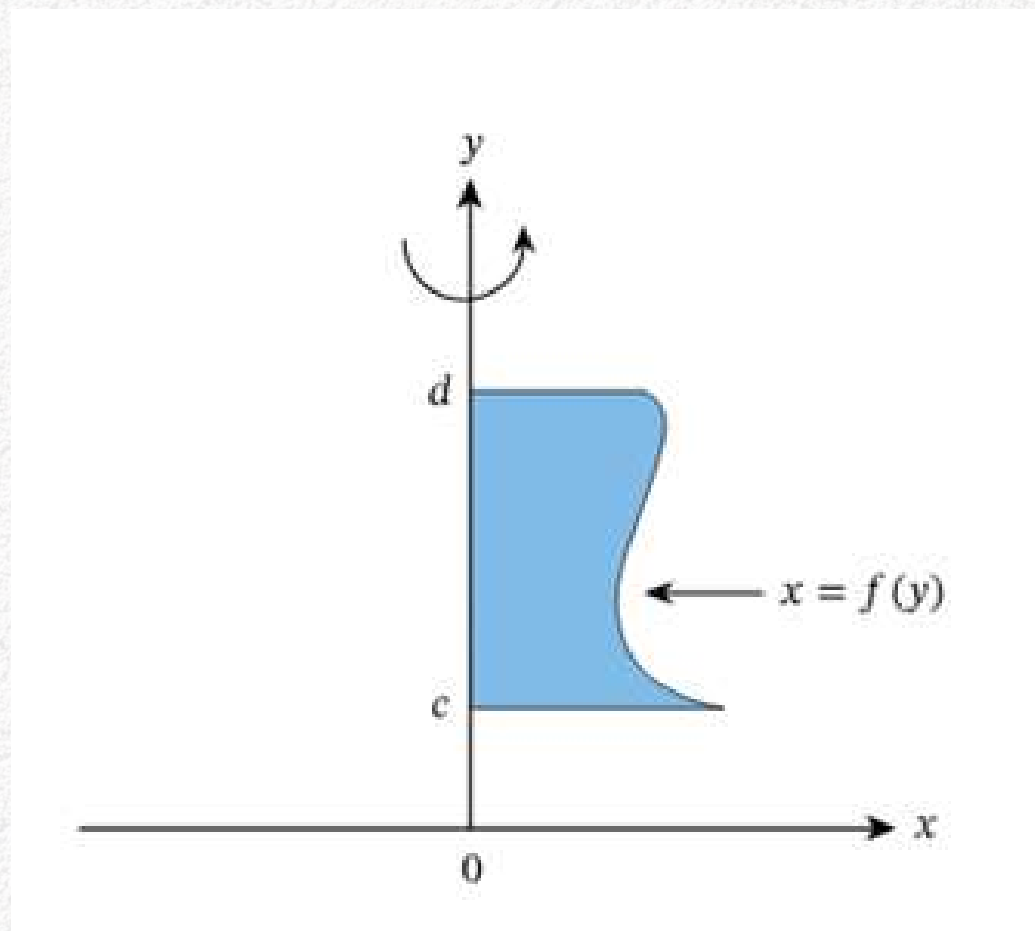
Find the volume of the solid obtained by rotating $y=x^2$ about the x-axis from $x=0$ to $x=1$:

$$V = \pi \int_0^1 (x^2)^2 dx$$

$$= \pi \int_0^1 x^4 dx$$

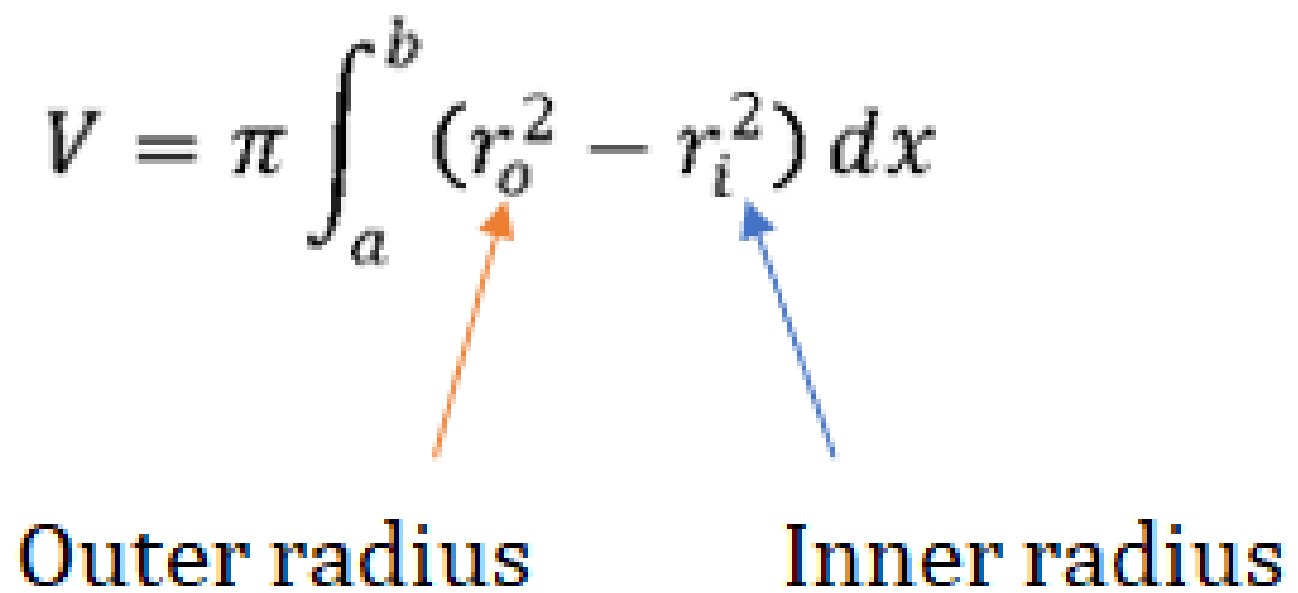
$$= \pi [5x]_0^1$$

$$= 5\pi$$



Washer Method:

In Washer Method we take the axis of revolution is not the boundary of the plane region and the cross section are taken perpendicular to the axis of revolution. Washer are considered to be the disk with hole. Now to calculate the area of the washer the formula is, $\pi(r_o^2 - r_i^2)$ and its volume is the area times its thickness.

$$V = \pi \int_a^b (r_o^2 - r_i^2) dx$$


Outer radius Inner radius

Example:

Find the volume of the solid obtained by rotating the region between $y=x^2$ and $y=x$ about the x -axis from $x=0$ to $x=1$:

$$V = \pi \int_0^1 [x^2 - (x^2)^2] dx$$

$$= \pi \int_0^1 [x^2 - x^4] dx$$

$$= \pi [3x^3 - 5x^5]_0^1$$

$$= 15\pi$$

Shell Method:

- Used for rotation about vertical or non-axis lines, where slicing the solid into cylinders (shells) is easier.

$$V = 2\pi \int_a^b (\text{radius}) \cdot (\text{height}) dx$$

where:

- Radius: Distance from the axis of rotation to the element.
- Height: The value of the function or the difference between functions.

THANK YOU

